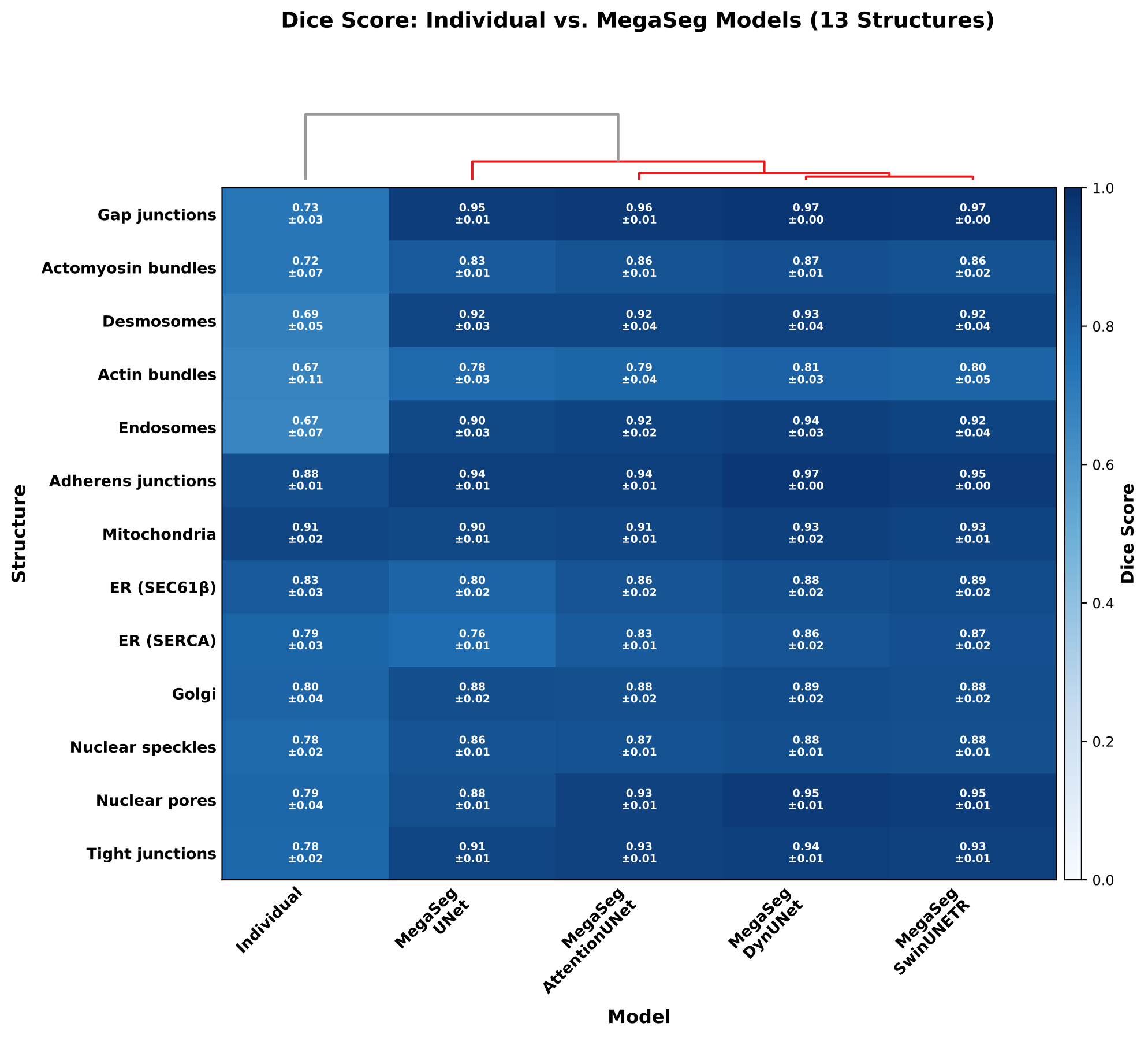
Dice Score: Individual vs. MegaSeg Models (13 Structures)
Structure
Model
Dice Score
1.0
0.8
0.6
0.4
0.2
0.0
Individual
MegaSeg UNet
MegaSeg AttentionUNet
MegaSeg DynUNet
MegaSeg SwinUNETR
Gap junctions
0.73 ±0.03
0.95 ±0.01
0.96 ±0.01
0.97 ±0.00
0.97 ±0.00
Actomyosin bundles
0.72 ±0.07
0.83 ±0.01
0.86 ±0.01
0.87 ±0.01
0.86 ±0.02
Desmosomes
0.69 ±0.05
0.92 ±0.03
0.92 ±0.04
0.93 ±0.04
0.92 ±0.04
Actin bundles
0.67 ±0.11
0.78 ±0.03
0.79 ±0.04
0.81 ±0.03
0.80 ±0.05
Endosomes
0.67 ±0.07
0.90 ±0.03
0.92 ±0.02
0.94 ±0.03
0.92 ±0.04
Adherens junctions
0.88 ±0.01
0.94 ±0.01
0.94 ±0.01
0.97 ±0.00
0.95 ±0.00
Mitochondria
0.91 ±0.02
0.90 ±0.01
0.91 ±0.01
0.93 ±0.02
0.93 ±0.01
ER (SEC61β)
0.83 ±0.03
0.80 ±0.02
0.86 ±0.02
0.88 ±0.02
0.89 ±0.02
ER (SERCA)
0.79 ±0.03
0.76 ±0.01
0.83 ±0.01
0.86 ±0.02
0.87 ±0.02
Golgi
0.80 ±0.04
0.88 ±0.02
0.88 ±0.02
0.89 ±0.02
0.88 ±0.02
Nuclear speckles
0.78 ±0.02
0.86 ±0.01
0.87 ±0.01
0.88 ±0.01
0.88 ±0.01
Nuclear pores
0.79 ±0.04
0.88 ±0.01
0.93 ±0.01
0.95 ±0.01
0.95 ±0.01
Tight junctions
0.78 ±0.02
0.91 ±0.01
0.93 ±0.01
0.94 ±0.01
0.93 ±0.01